



Probabilistische Datenwissenschaft für die Psychologie

Dirk Ostwald

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Welcome

Welcome to the working version of *Probabilistic Data Science for Psychology (PDSP)*, a textbook on data-scientific methodology at the Institute of Psychology at Otto von Guericke University Magdeburg.



Figure 1 Probabilistic Data Science for Psychology

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Part I

Mathematical Foundations

As the language of natural-scientific modeling, mathematics is also the language of probabilistic data science in psychology. In this role, mathematics mediates between intuitive scientific language and the variety of programming languages used to implement data-scientific analyses. Mathematical concept formation makes it possible to grasp concepts very precisely and in a generally understandable way, and to communicate them independently of the imprecision of everyday scientific language or the idiosyncrasies of different programming languages. Of course, mathematical language cannot assume this role detached from scientific-language intuition or practical application. In probabilistic data analysis, mathematics is therefore never an end in itself, but must always be understood as applied mathematics.

In this part, after some thoughts on the fundamental concepts of mathematics, we present sets and functions (Chapter 2), the two pillars of modern mathematics, in compressed form. We supplement this presentation with some notational aspects of sums, products, and powers. Closely interwoven with the concept of a function are differential calculus and integral calculus. Here, too, the aim is not an exhaustive presentation, but in particular an explanation of some foundations that are central to the data-scientific concepts of optimization and to working with probability functions. We supplement these sections with some introductory thoughts on the analytical foundations of differential and integral calculus. In dealing with the large data sets of contemporary science, matrix notation has proved useful. We first consider this subfield of linear algebra for single-column matrices and then for matrices in detail, again with the aim of providing notational and computational foundations for later sections.

Overall, this part therefore primarily serves as a brief introduction to foundations that will be taken up again in other parts, and in particular also as the introduction of a unified notation. Depending on need and interest, more in-depth self-study of the various contents is of course advisable. In the following literature notes, we provide an overview of sources and reading recommendations for the contents considered here.

Literature Notes

Bärwolff (2017) and Arens et al. (2018) form the primary basis for many of the topics presented here and offer an excellent and comprehensive entry point into the material covered. At the international level, Spivak (2008) and Strang (2009) provide very readable introductions to differential calculus and linear algebra, respectively. A deeper understanding, especially of analytical concepts, is provided, for example, by Abbott (2015) and Chiossi (2021). Searle (1982) offers a compact presentation of those aspects of matrix calculus and linear algebra that are used mainly in probabilistic data analysis. Deisenroth et al. (2020) provides an overview of many of the topics treated here under the moniker of machine learning.

1 Sums, Products, Powers

This unit introduces notation for the elementary arithmetic operations.

1.1 Sums

Definition 1.1 (Summation Symbol). It denotes

$$\sum_{i=1}^n x_i = x_1 + x_2 + \cdots + x_n. \quad (1.1)$$

Here,

- Σ denotes the Greek letter *Sigma*, mnemonically standing for *sum*,
- the subscript $i = 1$ denotes the *running index* and the *starting index*,
- the superscript n denotes the *terminal index*, and
- $x_1, x_2, \dots, x_n$ denote the *summands*.

•

For meaningful use of the summation symbol, it is essential to specify the beginning and the end of the summation by means of the subscript and the superscript. The precise name of the running index, by contrast, is irrelevant for the value of the sum; thus,

$$\sum_{i=1}^n x_i = \sum_{j=1}^n x_j. \quad (1.2)$$

Sometimes the running index is also given as an element of an *index set*. If, for example, the index set $I := \{1, 5, 7\}$ is defined, then

$$\sum_{i \in I} x_i := x_1 + x_5 + x_7. \quad (1.3)$$

In the following, we briefly consider a few examples of how the summation symbol is used.

- *Summation of predefined summands.* Let $x_1 := 2$, $x_2 := 10$, and $x_3 := -4$. Then

$$\sum_{i=1}^3 x_i = x_1 + x_2 + x_3 = 2 + 10 - 4 = 8. \quad (1.4)$$

- *Summation of weighted predefined summands.* Again let $x_1 := 2$, $x_2 := 10$, and $x_3 := -4$. In addition, let the *weighting coefficients* $a_1 := \frac{1}{2}$, $a_2 := \frac{1}{5}$, and $a_3 := 2$ be defined. Then

$$\sum_{i=1}^3 a_i x_i = a_1 x_1 + a_2 x_2 + a_3 x_3 = \frac{1}{2} \cdot 2 + \frac{1}{5} \cdot 10 + 2 \cdot (-4) = 1 + 2 - 8 = -5. \quad (1.5)$$

Expressions of the form $\sum_{i=1}^n a_i x_i$ are also called *linear combinations* of $x_1, \dots, x_n$ with the *coefficients* or *weighting parameters* $a_1, \dots, a_n$.

- *Summation of natural numbers.* We have

$$\sum_{i=1}^5 i = 1 + 2 + 3 + 4 + 5 = 15. \quad (1.6)$$

- *Summation of even natural numbers.* We have

$$\sum_{i=1}^5 2i = 2 \cdot 1 + 2 \cdot 2 + 2 \cdot 3 + 2 \cdot 4 + 2 \cdot 5 = 2 + 4 + 6 + 8 + 10 = 30. \quad (1.7)$$

- *Summation of odd natural numbers.* We have

$$\sum_{i=1}^5 (2i-1) = 2 \cdot 1 - 1 + 2 \cdot 2 - 1 + 2 \cdot 3 - 1 + 2 \cdot 4 - 1 + 2 \cdot 5 - 1 = 1 + 3 + 5 + 7 + 9 = 25. \quad (1.8)$$

Working with the summation symbol can often be simplified by applying the following calculation rules.

Theorem 1.1 (Calculation Rules for Sums).

- (1) *Sums of identical summands*

$$\sum_{i=1}^n x = nx \quad (1.9)$$

- (2) *Associativity for sums of equal length*

$$\sum_{i=1}^n x_i + \sum_{i=1}^n y_i = \sum_{i=1}^n (x_i + y_i) \quad (1.10)$$

- (3) *Distributivity under multiplication by a constant*

$$\sum_{i=1}^n a x_i = a \sum_{i=1}^n x_i \quad (1.11)$$

- (4) *Splitting sums for $1 < m < n$*

$$\sum_{i=1}^n x_i = \sum_{i=1}^m x_i + \sum_{i=m+1}^n x_i \quad (1.12)$$

- (5) *Reindexing*

$$\sum_{i=0}^n x_i = \sum_{j=m}^{n+m} x_{j-m} \quad (1.13)$$

◦

Proof. These calculation rules can be verified by writing out the sums and applying the rules of addition and multiplication. Here, we show associativity for sums of equal length and distributivity under multiplication by a constant as examples. For the former, we have

$$\begin{aligned}\sum_{i=1}^n x_i + \sum_{i=1}^n y_i &= x_1 + x_2 + \cdots + x_n + y_1 + y_2 + \cdots + y_n \\ &= x_1 + y_1 + x_2 + y_2 + \cdots + x_n + y_n \\ &= \sum_{i=1}^n (x_i + y_i).\end{aligned}\tag{1.14}$$

For the latter, we have

$$\begin{aligned}\sum_{i=1}^n ax_i &= ax_1 + ax_2 + \cdots + ax_n \\ &= a(x_1 + x_2 + \cdots + x_n) \\ &= a \sum_{i=1}^n x_i.\end{aligned}\tag{1.15}$$

□

Examples

As a first example of applying the calculation rules recorded in Theorem 1.1, we consider the evaluation of a *mean* (sometimes also called an *average*). Let $x_1, x_2, \dots, x_n$ be real numbers. The mean of these numbers is the sum of $x_1, x_2, \dots, x_n$ divided by the number of numbers n . By statement (3) of Theorem 1.1, it is irrelevant whether the numbers are first added up and the resulting sum is then divided by n , or whether the individual numbers are each divided by n and the corresponding results are then added up. More precisely, by applying Theorem 1.1 (3) with $a = 1/n$, we obtain

$$\frac{1}{n} \sum_{i=1}^n x_i = \sum_{i=1}^n \frac{x_i}{n}.\tag{1.16}$$

For example, the mean of $x_1 := 1, x_2 := 4, x_3 := 2$, and $x_4 := 1$ is given by

$$\frac{1}{4} \sum_{i=1}^4 x_i = \frac{1}{4}(1 + 4 + 2 + 1) = \frac{8}{4} = 2 = \frac{8}{4} = \frac{1}{4} + \frac{4}{4} + \frac{2}{4} + \frac{1}{4} = \sum_{i=1}^4 \frac{x_i}{4}.\tag{1.17}$$

As a second example, we consider the reindexing rule recorded in Theorem 1.1 (5). Let $n := 3$ and $m := 2$, and let $x_0 := 2, x_1 := 3, x_2 := 5$, and $x_3 := 10$. Then, evidently,

$$\begin{aligned}\sum_{i=0}^3 x_i &= x_0 + x_1 + x_2 + x_3 \\ &= 2 + 3 + 5 + 10 \\ &= 20.\end{aligned}\tag{1.18}$$

But we also have

$$\begin{aligned}
 \sum_{j=m}^{n+m} x_{j-m} &= \sum_{j=2}^{3+2} x_{j-2} \\
 &= \sum_{j=2}^5 x_{j-2} \\
 &= x_{2-2} + x_{3-2} + x_{4-2} + x_{5-2} \\
 &= x_0 + x_1 + x_2 + x_3 \\
 &= 2 + 3 + 5 + 10 \\
 &= 20.
 \end{aligned} \tag{1.19}$$

Double Sums

In applications, one often encounters expressions that carry out several summations one after another. For each of these sums, the definitions and calculation rules listed above apply. The meaning of double sums is best made clear by writing them out from the inside to the outside, while noting that the running index of the outer sum remains constant during each iteration of the inner sum.

The following examples may illustrate this.

(1)

$$\begin{aligned}
 \sum_{i=1}^2 \sum_{j=1}^3 (i+j) &= \sum_{i=1}^2 (i+1 + i+2 + i+3) \\
 &= (1+1 + 1+2 + 1+3) + (2+1 + 2+2 + 2+3) \\
 &= 9 + 12 \\
 &= 21
 \end{aligned} \tag{1.20}$$

(2)

$$\begin{aligned}
 \sum_{i=1}^2 \sum_{j=1}^3 (x_i + y_j) &= \sum_{i=1}^2 (x_i + y_1 + x_i + y_2 + x_i + y_3) \\
 &= (x_1 + y_1 + x_1 + y_2 + x_1 + y_3) + (x_2 + y_1 + x_2 + y_2 + x_2 + y_3)
 \end{aligned} \tag{1.21}$$

(3)

$$\begin{aligned}
 \sum_{i=1}^3 \sum_{j=1}^2 (x_i y_j) &= \sum_{i=1}^3 (x_i y_1 + x_i y_2) \\
 &= (x_1 y_1 + x_1 y_2) + (x_2 y_1 + x_2 y_2) + (x_3 y_1 + x_3 y_2)
 \end{aligned} \tag{1.22}$$

1.2 Products

The product symbol provides notation for products that is analogous to the summation symbol.

Definition 1.2 (Product Symbol). It denotes

$$\prod_{i=1}^n x_i = x_1 \cdot x_2 \cdot \cdots \cdot x_n. \quad (1.23)$$

Here,

- $\prod$ denotes the Greek letter *Pi*, mnemonically standing for *product*,
- the subscript $i = 1$ denotes the *running index* and the *starting index*,
- the superscript n denotes the *terminal index*, and
- $x_1, x_2, \dots, x_n$ denote the *product terms*.

•

Analogously to the summation symbol, the product symbol is meaningful only with a subscript and a superscript specifying the running and terminal index. The precise name of the running index is again irrelevant; thus,

$$\prod_{i=1}^n x_i = \prod_{j=1}^n x_j. \quad (1.24)$$

Here, too, the running index is in rare cases given as an element of an index set. If, for example, the index set $J := \mathbb{N}_2^0$ is defined, then

$$\prod_{j \in J} x_j := x_0 \cdot x_1 \cdot x_2. \quad (1.25)$$

One example of the use of the product symbol is the definition of the *factorial* of a natural number n by

$$n! := \prod_{i=1}^n i. \quad (1.26)$$

For example,

$$3! := \prod_{i=1}^3 i = 1 \cdot 2 \cdot 3 = 6. \quad (1.27)$$

There are also a number of calculation rules for products that often simplify working with them. We list some in the following theorem. In doing so, we make anticipatory use of the notation of *powers* defined in Section 1.3.

Theorem 1.2 (Calculation Rules for Products).

(1) *Products of identical factors*

$$\prod_{i=1}^n x = x^n \quad (1.28)$$

(2) *Raising constants to powers*

$$\prod_{i=1}^n ax_i = a^n \prod_{i=1}^n x_i \quad (1.29)$$

(3) *Splitting products for $1 < m < n$*

$$\prod_{i=1}^n x_i = \prod_{i=1}^m x_i \prod_{j=m+1}^n x_j \quad (1.30)$$

(4) *Product of products*

$$\prod_{i=1}^n x_i y_i = \prod_{i=1}^n x_i \prod_{i=1}^n y_i \quad (1.31)$$

◦

1.3 Powers

Products of numbers with themselves can be abbreviated using power notation.

Definition 1.3 (Power). For $a \in \mathbb{R}$ and $n \in \mathbb{N}^0$, the n -th power of a is defined by

$$a^0 := 1 \text{ and } a^{n+1} := a^n \cdot a. \quad (1.32)$$

Furthermore, for $a \in \mathbb{R} \setminus \{0\}$ and $n \in \mathbb{N}^0$, the *negative n -th power of a* is defined by

$$a^{-n} := (a^n)^{-1} := \frac{1}{a^n}. \quad (1.33)$$

a is called the *base*, and n is called the *exponent*.

•

The way a^{n+1} is defined by referring back to the power a^n in the above definition is called *recursive*. The definition $a^0 := 1$ is called the *initial case* of the recursion; it is what makes the recursive definition of a^{n+1} possible in the first place. The definition $a^{n+1} := a^n \cdot a$ is also called the *recursion step*. The following calculation rules simplify computations with powers.

Theorem 1.3 (Calculation Rules for Powers). For $a, b \in \mathbb{R}$ and $n, m \in \mathbb{Z}$ with $a \neq 0$ for negative exponents, the following calculation rules hold:

$$a^n a^m = a^{n+m} \quad (1.34)$$

$$(a^n)^m = a^{nm} \quad (1.35)$$

$$(ab)^n = a^n b^n \quad (1.36)$$

◦

Instead of a proof, we consider the following three examples.

(1)

$$2^2 \cdot 2^3 = (2 \cdot 2) \cdot (2 \cdot 2 \cdot 2) = 2^5 = 2^{2+3} \quad (1.37)$$

(2)

$$(3^2)^3 = (3 \cdot 3)^3 = (3 \cdot 3) \cdot (3 \cdot 3) \cdot (3 \cdot 3) = 3^6 = 3^{2 \cdot 3} \quad (1.38)$$

(3)

$$(2 \cdot 4)^2 = (2 \cdot 4) \cdot (2 \cdot 4) = (2 \cdot 2) \cdot (4 \cdot 4) = 2^2 \cdot 4^2 \quad (1.39)$$

Closely related to powers is the definition of the n -th root:

Definition 1.4 (n -th Root). For $a \in \mathbb{R}_{\geq 0}$ and $n \in \mathbb{N}$, the n -th root of a is defined as the nonnegative real number r with

$$r^n = a. \quad (1.40)$$

•

When computing with roots, power notation for roots is often helpful because it allows the calculation rules for powers to be applied directly.

Theorem 1.4 (Power Notation for the n -th Root). Let $a \in \mathbb{R}_{\geq 0}$, let $n \in \mathbb{N}$, and let r be the n -th root of a . Then

$$r = a^{\frac{1}{n}} \quad (1.41)$$

◦

Proof. We have

$$\left(a^{\frac{1}{n}}\right)^n = a^{\frac{1}{n}} \cdot a^{\frac{1}{n}} \cdot \dots \cdot a^{\frac{1}{n}} = a^{\sum_{i=1}^n \frac{1}{n}} = a^1 = a. \quad (1.42)$$

Thus, by Definition 1.4, $r = a^{\frac{1}{n}}$. □

Computing with square roots is greatly facilitated by the power notation

$$\sqrt{x} = x^{\frac{1}{2}} \quad (1.43)$$

For example,

$$\frac{2\pi}{\sqrt{2\pi}} = \frac{2\pi}{(2\pi)^{\frac{1}{2}}} = (2\pi)^1 \cdot (2\pi)^{-\frac{1}{2}} = (2\pi)^{1-\frac{1}{2}} = (2\pi)^{\frac{1}{2}} = \sqrt{2\pi}. \quad (1.44)$$

The following relation between square, root, and absolute value is used quite often.

Theorem 1.5 (Root and Absolute Value). Let $x \in \mathbb{R}$, and let

$$|x| := \begin{cases} x & \text{for } x \geq 0 \\ -x & \text{for } x < 0 \end{cases} \quad (1.45)$$

denote the absolute value of x . Then

$$\sqrt{x^2} = |x|. \quad (1.46)$$

◦

Proof. We consider the cases $x \geq 0$ and $x < 0$. First, let $x \geq 0$. Then

$$x \geq 0 \Rightarrow x^2 \geq 0 \Rightarrow \sqrt{x^2} = x = |x|. \quad (1.47)$$

Now let $x < 0$. Then

$$x < 0 \Rightarrow x^2 > 0 \Rightarrow \sqrt{x^2} = -x = |x|. \quad (1.48)$$

□

2 Functions

Alongside sets, functions are the second foundational pillar of modern mathematics. In this unit, we define the concept of a function, introduce first properties of functions, and give an overview of several elementary functions.

2.1 Definition and Properties

Definition 2.1 (Function). A *function* or *mapping* f is an assignment rule that assigns exactly one element of a set Z to each element of a set D . D is called the *domain* of f , and Z is called the *codomain* of f . We write

$$f : D \rightarrow Z, x \mapsto f(x), \quad (2.1)$$

where $f : D \rightarrow Z$ is read as “the function f maps all elements of the set D uniquely to elements in Z ” and $x \mapsto f(x)$ is read as “ x , which is an element of D , is mapped by the function f to $f(x)$, where $f(x)$ is an element of Z .” The arrow $\rightarrow$ denotes the mapping between the sets D and Z , while the arrow $\mapsto$ denotes the mapping between an element of D and an element of Z .

•

It is essential to distinguish between the *function* f as an assignment rule and a *value of the function* $f(x)$ as an element of Z . x is the *argument* of the function (the function’s *input*), and $f(x)$ is the value that the function f takes for the argument x (the function’s *output*). Usually, the definition of a function specifies, after $f(x)$, the *functional form of f* , that is, a rule for forming the value $f(x)$ from x . For example, in the following definition of a function

$$f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}, x \mapsto f(x) := x^2 \quad (2.2)$$

the definition of a power is used. Note that functions are always *unique* in the sense that, whenever the function is applied, they always assign one and the same $f(x) \in Z$ to each $x \in D$.

Figure 2.1 visualizes several aspects of the definition of a function. Figure 2.1 A first illustrates the central terms of the definition. Figure 2.1 B shows a first example of a function in which the function values assigned to the arguments are determined not by a computational rule but by direct definition. Figure 2.1 C and Figure 2.1 D use the same pictorial language to emphasize two aspects of the definition by means of counterexamples: the drawing in Figure 2.1 C is not the representation of a function, because by Definition 2.1 a function assigns *each element* of a set D exactly one element of a codomain Z . The sketched assignment rule does not assign any element in Z to the element $2 \in D$, and therefore it is not a function. Similarly, according to Definition 2.1, a function assigns exactly one element of a codomain Z to each element of a set D . The assignment rule

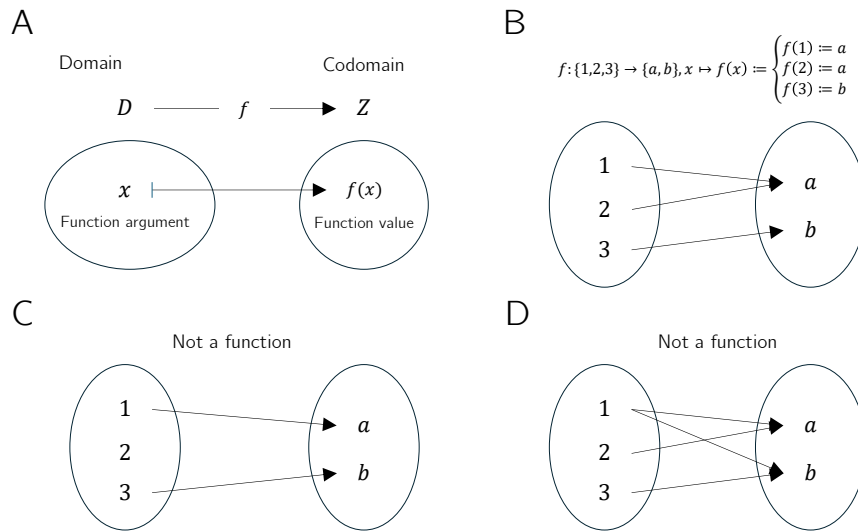


Figure 2.1 Aspects of the definition of a function

sketched in Figure 2.1 D assigns both the element $a \in Z$ and the element $b \in Z$ to $1 \in D$, and therefore it is incompatible with Definition 2.1.

Thus, functions put elements of sets into relation with one another. The sets of these elements receive special names.

Definition 2.2 (Image, Range, Preimage Set, Preimage). Let $f: D \rightarrow Z, x \mapsto f(x)$ be a function and let $D' \subseteq D$ and $Z' \subseteq Z$. The set

$$f(D') := \{z \in Z \mid \text{there exists an } x \in D' \text{ with } z = f(x)\} \quad (2.3)$$

is called the *image of D'* , and $f(D) \subseteq Z$ is called the *range of f* . Furthermore, the set

$$f^{-1}(Z') := \{x \in D \mid f(x) \in Z'\} \quad (2.4)$$

is called the *preimage set of Z'* . An $x \in D$ with $z = f(x) \in Z$ is called a *preimage of z* .

•

Note that the range $f(D)$ of f and the codomain Z of f need not be identical.

Example

To clarify the concepts introduced in Definition 2.2, we consider the function shown in Figure 2.2 A,

$$f: \{1, 2, 3, 4, 5\} \rightarrow \{a, b, c, d\}, x \mapsto f(x) := \begin{cases} f(1) := b \\ f(2) := d \\ f(3) := c \\ f(4) := c \\ f(5) := d \end{cases} \quad (2.5)$$

According to Definition 2.2, an image is always defined with respect to a subset D' of the domain D . Thus, let $D' := \{2, 3\} \subset D$, as shown in Figure 2.2 B. Then the image

of D' is the set of those $z \in Z$ for which there exists an $x \in D'$ such that $z = f(x)$. Here, the relevant $z \in Z$ for which such an $x \in D'$ exists are precisely $c, d \in Z$, because $f(2) = d$ and $f(3) = c$. Beyond these, there is no $z \in Z$ with $f(x) = z$ and $x \in \{2, 3\}$. By Definition 2.2, the range $f(D)$ is the subset of those $z \in Z$ for which there exists an $x \in D$ such that $z = f(x)$. This is true for all elements of Z except for $a \in Z$, because there is no $x \in D$ with $a = f(x)$. For the function considered here, the range of f is therefore given by $f(D) := \{b, c, d\}$.

Conversely, according to Definition 2.2, a preimage set is always defined with respect to a subset Z' of the codomain Z . Thus, let $Z' := \{c, d\} \subset Z$, as shown in Figure 2.2 C. Then the preimage set of Z' is the set of those $x \in D$ for which $f(x) \in Z'$. The elements of D whose function values under f are elements of Z' are precisely $\{2, 3, 4, 5\}$. By contrast, $1 \in D$ is not an element of this preimage set, because f maps $1 \in D$ to $b \in Z$ and $b \notin Z'$. Nevertheless, $1 \in D$ is of course a preimage of b .

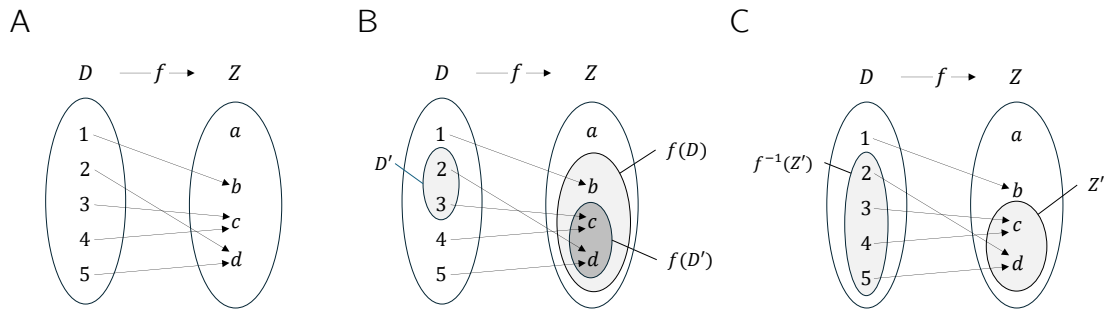


Figure 2.2 (A) Visualization of the function Equation 2.5. (B) Range and an image of f . (C) Preimage set of a set Z' under f

Basic properties of functions are named in the following definition.

Definition 2.3 (Injectivity, Surjectivity, Bijectivity). Let $f : D \rightarrow Z, x \mapsto f(x)$ be a function. f is called *injective* if, for every image element $z \in f(D)$, there is exactly one preimage $x \in D$. Equivalently, f is injective if $x_1, x_2 \in D$ and $x_1 \neq x_2$ imply $f(x_1) \neq f(x_2)$. f is called *surjective* if $f(D) = Z$, that is, if every element of the codomain Z has a preimage in the domain D . Finally, f is called *bijective* if it is both injective and surjective. Bijective functions are also called *one-to-one correspondences* or *one-to-one mappings*.

•

Examples

We first illustrate Definition 2.3 using three (counter)examples in Figure 2.3. Figure 2.3 A shows the *non-injective* function

$$f : \{1, 2, 3\} \rightarrow \{a, b\}, x \mapsto f(x) := \begin{cases} f(1) := a \\ f(2) := a \\ f(3) := b \end{cases} . \quad (2.6)$$

The function is non-injective because the element a in the range of f has more than one preimage in the domain of f , namely the elements 1 and 2. Figure 2.3 B shows the

non-surjective function

$$g : \{1, 2, 3\} \rightarrow \{a, b, c, d\}, x \mapsto g(x) := \begin{cases} g(1) := a \\ g(2) := b \\ g(3) := d \end{cases} . \quad (2.7)$$

The function is non-surjective because the element c in the codomain of g has no preimage in the domain of g . Finally, Figure 2.3 C shows the bijective function

$$h : \{1, 2, 3\} \rightarrow \{a, b, c\}, x \mapsto h(x) := \begin{cases} h(1) := a \\ h(2) := b \\ h(3) := c \end{cases} . \quad (2.8)$$

For *every* element in the codomain of h there is *exactly one* preimage, so the function is injective and surjective and therefore bijective.

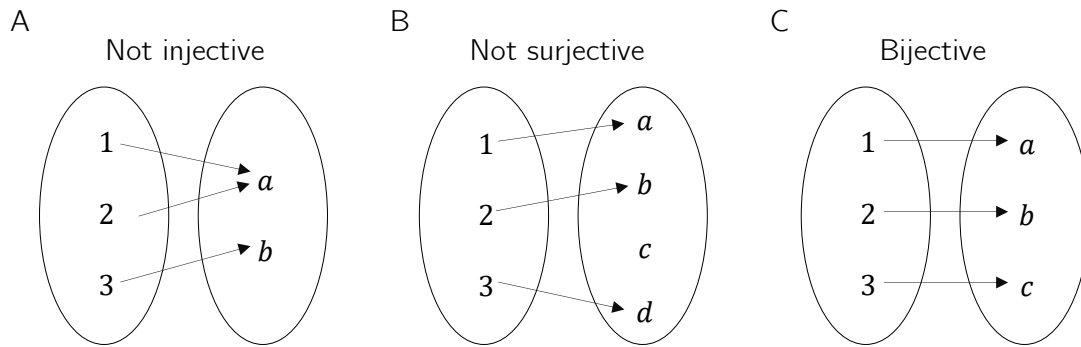


Figure 2.3 Injectivity, surjectivity, bijectivity.

As a further example, consider the function

$$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto f(x) := x^2 \quad (2.9)$$

This function is not injective because, for example, with $x_1 = 2 \neq -2 = x_2$ we have $f(x_1) = 2^2 = 4 = (-2)^2 = f(x_2)$. Moreover, f is not surjective because, for example, $-1 \in \mathbb{R}$ has no preimage under f . However, if the domain of f is restricted to the non-negative real numbers, that is, if we define the function

$$\tilde{f} : [0, \infty[\rightarrow [0, \infty[, x \mapsto \tilde{f}(x) := x^2, \quad (2.10)$$

then, in contrast to f , $\tilde{f}$ is injective and surjective, hence bijective.

On the uncountability of the real numbers

At this point, we use the concepts of surjectivity and bijectivity of functions to deepen the idea of the *uncountability of the real numbers* a little. This idea says that there are different kinds of infinities, which is certainly not easy to grasp intuitively at first. Moreover, some aspects of probability theory are formulated in ways that accommodate the uncountability of the real numbers, so even a rough understanding of this property can help motivate some of the subtleties of probability theory. Specifically, we briefly sketch *Cantor's diagonal argument* (Cantor (1891), Gray (1994)) for the uncountability of the real numbers. We begin by recording what, against the background of bijective mappings, is meant by a *finite* and a *countably infinite* set.

Definition 2.4. A set M is called *finite* if there exists a bijective mapping of the form

$$f : \mathbb{N}_n \rightarrow M \quad (2.11)$$

onto the elements of M . Furthermore, a set M is called *countably infinite* if there exists a bijective mapping of the form

$$f : \mathbb{N} \rightarrow M \quad (2.12)$$

onto the elements of M .

•

In the case of a finite set, each element of the set can thus be assigned a natural number with maximum value $n < \infty$ in the sense of a bijective mapping. In the case of a countably infinite set, each element of the set can at least be assigned a natural number in the sense of a bijective mapping. The elements of these sets can therefore be counted.

Cantor (1891) showed that this is not true in general for the real numbers: no bijective mapping between the real numbers and the natural numbers can be constructed, and hence there must be more real numbers than natural numbers. More specifically, one can argue that even the interval $[0, 1]$ already contains more real numbers than there are natural numbers, that is, no surjective function from $\mathbb{N}$ to $[0, 1]$ can be constructed.

Cantor's argument has the following form. Suppose that $f : \mathbb{N} \rightarrow [0, 1]$ is an injective function represented in tabular form, for example as in Table 2.1.

Table 2.1 Mapping $f : \mathbb{N} \rightarrow [0, 1]$. The dots $\dots$ and $:$ indicate that the table continues indefinitely to the right and downward.

n	$f(n)$										
1	0.	3	1	4	1	5	9	2	6	5	...
2	0.	8	9	4	5	9	7	8	1	0	...
3	0.	9	6	2	3	2	1	5	8	7	...
4	0.	5	3	6	8	8	8	9	7	3	...
5	0.	7	4	3	8	1	8	0	9	7	...
$:$	$:$										

Now construct another number in $[0, 1]$ by adding 1 to each of the bold diagonal entries in Table 2.1 whenever the corresponding value is less than 9, and otherwise setting the value to 0. This is shown in Table 2.2.

Table 2.2 Construction of another real number in $[0, 1]$.

n	$f(n)$										
1	0.	4	1	4	1	5	9	2	6	5	...
2	0.	8	0	4	5	9	7	8	1	0	...
3	0.	9	6	3	3	2	1	5	8	7	...
4	0.	5	3	6	9	8	8	9	7	3	...
5	0.	7	4	3	8	2	8	0	9	7	...
$:$	$:$										

The number thus constructed,

$$0.40392... \quad (2.13)$$

cannot occur in Table 2.1, because it differs from $f(1)$ in its first decimal place, from $f(2)$ in its second decimal place, from $f(3)$ in its third decimal place, ..., from $f(n)$ in its n th decimal place, and so on indefinitely. Thus there is at least one number in $[0, 1]$ to which no natural number can be assigned. The injective mapping f is not surjective and therefore not bijective. Consequently, $[0, 1]$ is not countably infinite and is accordingly called *uncountable*.

2.2 Types of Functions

By composition, further functions can be formed from given functions.

Definition 2.5 (Composition of Functions). Let $f : D \rightarrow Z$ and $g : Z \rightarrow S$ be two functions, where the range of f is assumed to agree with the domain of g . Then

$$g \circ f : D \rightarrow S, x \mapsto (g \circ f)(x) := g(f(x)) \quad (2.14)$$

defines a function called the *composition of f and g* .

•

The notation for composed functions takes a little getting used to. It is important to recognize that $g \circ f$ denotes the composed function and $(g \circ f)(x)$ denotes an element in the codomain of the composed function. Intuitively, when evaluating $(g \circ f)(x)$, one first applies the function f to x and then applies the function g to the element $f(x)$ of Z . This is captured by the functional form $g(f(x))$. For simplicity, the composition of two functions is often denoted by a single letter; for example, one writes $h := g \circ f$ with $h(x) = g(f(x))$. A mild source of confusion can arise when elements in the codomain of f are denoted by y , so that the notation $y = f(x)$ and $h(x) = g(y)$ is used. However, this notation is sometimes needed for notational simplicity. We visualize Definition 2.5 in Figure 2.4.

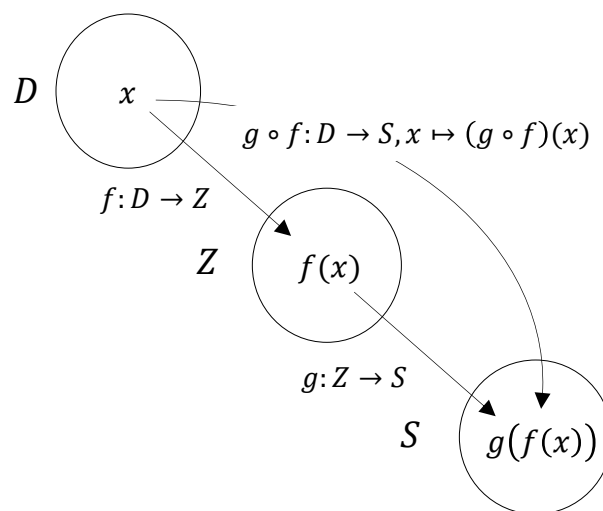


Figure 2.4 Composition of functions.

As an example of the composition of two functions, consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto f(x) := -x^2 \quad (2.15)$$

and

$$g : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto g(x) := \exp(x). \quad (2.16)$$

In this case, the composition of f and g is

$$g \circ f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto (g \circ f)(x) := g(f(x)) = \exp(-x^2). \quad (2.17)$$

A first application of function composition appears in the following definition.

Definition 2.6 (Inverse Function). Let $f : D \rightarrow Z, x \mapsto f(x)$ be a bijective function. Then the function f^{-1} with

$$f^{-1} \circ f : D \rightarrow D, x \mapsto (f^{-1} \circ f)(x) := f^{-1}(f(x)) = x \quad (2.18)$$

is called the *inverse function*, *inverse mapping*, or simply the *inverse of f* .

•

Inverse functions are always bijective. This follows because f is bijective and therefore each $x \in D$ is assigned exactly one $f(x) = z \in Z$. Hence, each $z \in Z$ is also assigned exactly one $x \in D$, namely $f^{-1}(f(x)) = x$. Intuitively, the inverse function of f undoes the effect of f on an element x . We visualize Definition 2.6 in Figure 2.5 A.

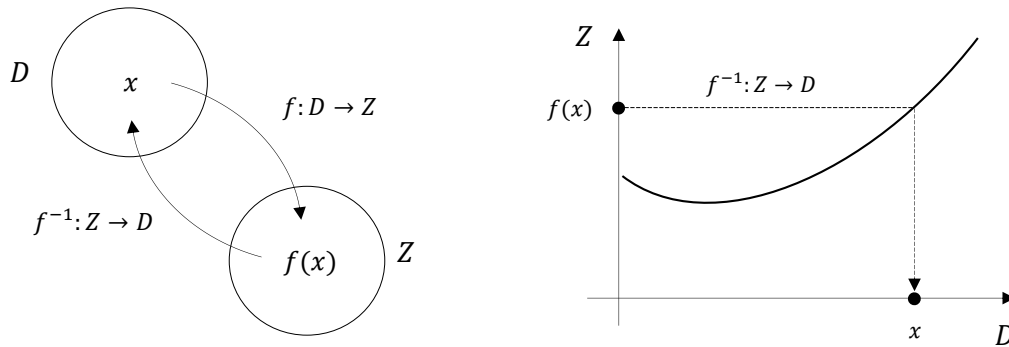


Figure 2.5 Inverse function.

If one considers the graph of a function in a Cartesian coordinate system, then applying a function to a value on the x -axis leads to a value on the y -axis. Applying the inverse function correspondingly leads from a value on the y -axis to a value on the x -axis. We visualize this in Figure 2.5 B. For example, consider the function

$$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto f(x) := 2x =: y. \quad (2.19)$$

Then the inverse function of f is given by

$$f^{-1} : \mathbb{R} \rightarrow \mathbb{R}, y \mapsto f^{-1}(y) := \frac{1}{2}y, \quad (2.20)$$

because for every $x \in \mathbb{R}$,

$$(f^{-1} \circ f)(x) := f^{-1}(f(x)) = f^{-1}(2x) = \frac{1}{2} \cdot 2x = x. \quad (2.21)$$

An important class of functions consists of *linear maps*.

Definition 2.7 (Linear Map). A map $f : D \rightarrow Z, x \mapsto f(x)$ is called a *linear map* if, for $x, y \in D$ and a scalar c ,

$$f(x + y) = f(x) + f(y) \quad (\text{additivity})$$

and

$$f(cx) = cf(x) \quad (\text{homogeneity})$$

hold. A map for which the above properties do not hold is called a *nonlinear map*. •

Linear maps are often known as “straight lines.” The general definition of linear maps is not completely congruent with this intuition. In particular, linear maps are only those functions that map zero to zero. We show this in the following theorem.

Theorem 2.1 (Linear Map of Zero). *Let $f : D \rightarrow Z$ be a linear map. Then*

$$f(0) = 0. \quad (2.22)$$

◦

Proof. First note that by additivity of f ,

$$f(0) = f(0 + 0) = f(0) + f(0). \quad (2.23)$$

Adding $-f(0)$ to both sides of the above equation gives

$$f(0) - f(0) = f(0) + f(0) - f(0) \Leftrightarrow f(0) = 0 \quad (2.24)$$

and this proves the claim. □

We illustrate the concept of a linear map with two examples.

- For $a \in \mathbb{R}$, the map

$$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto f(x) := ax \quad (2.25)$$

is a linear map because

$$f(x + y) = a(x + y) = ax + ay = f(x) + f(y) \text{ and } f(cx) = acx = cax = cf(x). \quad (2.26)$$

- For $a, b \in \mathbb{R}$, by contrast, the map

$$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto f(x) := ax + b \quad (2.27)$$

is nonlinear because, for example, for $a := b := 1$,

$$f(x + y) = 1(x + y) + 1 = x + y + 1 \neq x + 1 + y + 1 = f(x) + f(y). \quad (2.28)$$

A map of the form $f(x) := ax + b$ is called a *linear-affine map* or *linear-affine function*. Somewhat imprecisely, functions of the form $f(x) := ax + b$ are sometimes also called *linear functions*.

Besides the types of functions discussed so far, there are many further classes of functions. In the following definition, we classify functions by the dimensionality of their domains and codomains. This type of function classification is often helpful for gaining an initial overview of a mathematical model.

Definition 2.8 (Function Types). We distinguish

- *univariate real-valued functions* of the form

$$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto f(x), \quad (2.29)$$

- *multivariate real-valued functions* of the form

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, x \mapsto f(x) = f(x_1, \dots, x_n), \quad (2.30)$$

- and *multivariate vector-valued functions* of the form

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m, x \mapsto f(x) = \begin{pmatrix} f_1(x_1, \dots, x_n) \\ \vdots \\ f_m(x_1, \dots, x_n) \end{pmatrix}, \quad (2.31)$$

where $f_i, i = 1, \dots, m$ are called the *components* or *component functions* of f .

•

In physics, multivariate real-valued functions are called *scalar fields*, and multivariate vector-valued functions are called *vector fields*. In some applications, *matrix-variate matrix-valued functions* also occur.

2.3 Elementary Functions

By *elementary functions* we mean a small set of univariate real-valued functions that frequently appear as building blocks of more complex functions. These are the *polynomial functions*, the *exponential function*, the *logarithm function*, and the *gamma function*. In the following, we give the definitions of these functions and state their main properties as theorems, and then present their graphs. For proofs of the properties introduced here, we refer to the advanced literature.

Definition 2.9 (Polynomial Functions). A function of the form

$$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto f(x) := \sum_{i=0}^k a_i x^i = a_0 + a_1 x^1 + a_2 x^2 + \dots + a_k x^k \quad (2.32)$$

is called a *polynomial function* of degree k with coefficients $a_0, a_1, \dots, a_k \in \mathbb{R}$. Typical polynomial functions are listed in Table 2.3.

Table 2.3 Polynomial functions.

Name	Form	Coefficients
Constant function	$f(x) = a$	$a_0 := a, a_i := 0, i > 0$
Identity function	$f(x) = x$	$a_0 := 0, a_1 := 1, a_i := 0, i > 1$
Linear-affine function	$f(x) = ax + b$	$a_0 := b, a_1 := a, a_i := 0, i > 1$
Square function	$f(x) = x^2$	$a_0 := 0, a_1 := 0, a_2 := 1, a_i := 0, i > 2$

•

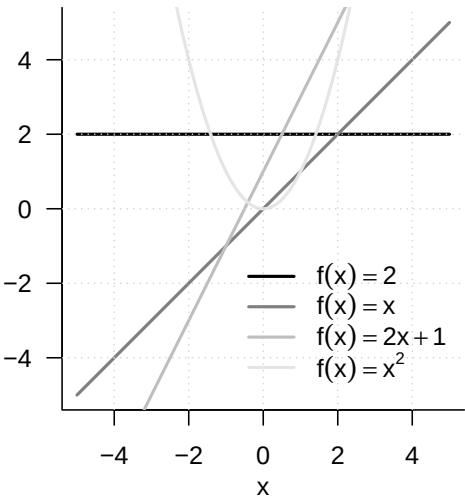


Figure 2.6 Selected polynomial functions

Figure 2.6 shows the graphs of the polynomial functions listed in Definition 2.9.

An important pair of functions consists of the exponential function and the logarithm function.

Theorem 2.2 (Exponential Function and Its Properties). *The exponential function is defined as*

$$\exp : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \exp(x) := e^x := \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

(2.33)

Among others, the exponential function has the properties listed in Table 2.4.

Table 2.4 Properties of the exponential function.

Property	Meaning
Range	$x \in]-\infty, 0[\Rightarrow \exp(x) \in]0, 1[$ $x \in]0, \infty[\Rightarrow \exp(x) \in]1, \infty[$
Monotonicity	$x < y \Rightarrow \exp(x) < \exp(y)$
Special values	$\exp(0) = 1$ and $\exp(1) = e$
Summation property	$\exp(x + y) = \exp(x) \exp(y)$
Subtraction property	$\exp(x - y) = \frac{\exp(x)}{\exp(y)}$

◦

In particular, the exponential function only takes positive values and intersects the y -axis at $x = 0$. The number $\exp(1) := e \approx 2.71\dots$ is called *Euler’s number*. Finally, the special values of the exponential function imply

$$\exp(x) \exp(-x) = \exp(x - x) = \exp(0) = 1.$$

(2.34)

Theorem 2.3 (Logarithm Function and Its Properties). *The logarithm function is defined as the inverse function of the exponential function,*

$$\ln :]0, \infty[\rightarrow \mathbb{R}, x \mapsto \ln(x) \text{ with } \ln(\exp(x)) = x \text{ for all } x \in \mathbb{R}.$$

(2.35)

Among others, the logarithm function has the properties listed in Table 2.5.

Table 2.5 Properties of the logarithm function.

Property	Meaning
Range	$x \in]0, 1[\Rightarrow \ln(x) \in]-\infty, 0[$ $x \in]1, \infty[\Rightarrow \ln(x) \in]0, \infty[$
Monotonicity	$x < y \Rightarrow \ln(x) < \ln(y)$
Special values	$\ln(1) = 0$ and $\ln(e) = 1$
Product property	$\ln(xy) = \ln(x) + \ln(y)$
Power property	$\ln(x^c) = c \ln(x)$
Division property	$\ln\left(\frac{1}{x}\right) = -\ln(x)$

◦

In contrast to the exponential function, the logarithm function takes both negative and positive values. The logarithm function intersects the x -axis at $x = 1$. The product property and the power property are central when calculating with the logarithm function. Intuitively, they can be remembered as: “The logarithm function turns products into sums and powers into products.” The graphs of the exponential and logarithm functions are shown in Figure 2.7.

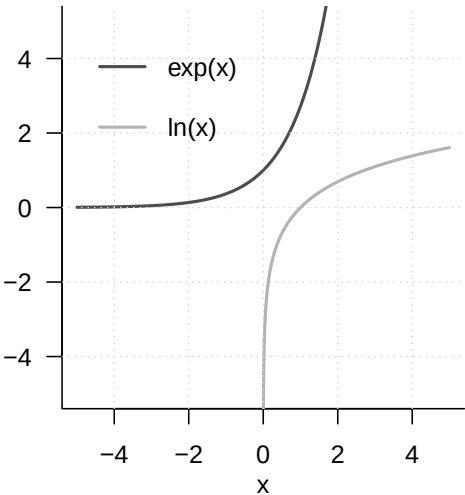


Figure 2.7 Exponential function and logarithm function

A frequent companion in probability theory is the *gamma function*.

Definition 2.10 (Gamma Function). The *gamma function* is defined by

$$\Gamma : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \Gamma(x) := \int_0^\infty \xi^{x-1} \exp(-\xi) d\xi$$

(2.36)

Among others, the gamma function has the properties listed in Table 2.6.

Table 2.6 Properties of the gamma function.

Property	Meaning
Special values	$\Gamma(1) = 1$
	$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$
	$\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$
Recursion property	For $x > 0$, $\Gamma(x+1) = x\Gamma(x)$

•

A section of the graph of the gamma function is shown in Figure 2.8.

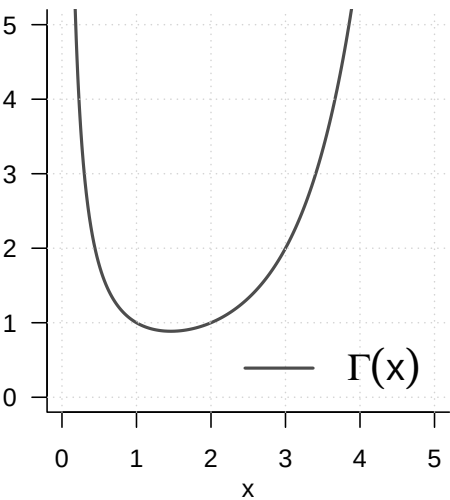


Figure 2.8 Gamma function

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